

SSIE-500: Homework 6

Measuring Complexity of Languages (2)

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Deadline: November 24, 2025 (1:30pm Eastern Time)

1. Read in a sufficiently long text document in English. Calculate the information entropy of the frequency distribution of characters that appear in the text. You can keep spaces, punctuations, and other symbols in this calculation too.
2. Write a computer program that generates a Huffman code for a given data set. Then use the generated Huffman code to encode the text from Step 1 into binary (0/1) strings.
3. Calculate the average code word length (i.e., the average number of bits per character in the original text data) achieved by the Huffman code, and compare it with the information entropy in the text calculated above.
4. Use the Huffman code generated above to encode *sentences* in the document (i.e., not the entire document) into binary strings, and calculate how many bits are spent per sentence *on average*.
5. Repeat Steps 1-4 for two other human languages, and compare the final results (average number of bits per sentence) across languages. What do you find?

Produce a PDF file of your report using L^AT_EX, showing your program code, results, and discussions of your findings. Submit it to the Assignment page of Brightspace by the deadline.